

- Q.24 What is the disadvantages of pre-stressed concrete (CO1)
- Q.25 Explain pre-tensioning method with diagram. (CO1)
- Q.26 Comparison between pre-tensioning and post-tensioning method. (CO1)
- Q.27 Explain concrete intruded piles. (CO2)
- Q.28 Write a short note on pre-cast piles. (CO2)
- Q.29 Explain pile foundation. (CO2)
- Q.30 Explain different types of classes that occur is pre-stressed Concrete. (CO1)
- Q.31 How anchorage can be protected from corrosion. (CO1)
- Q.32 Differentiate between shrinkage and creep of concrete. (CO1)
- Q.33 Discuss loss due elastic deformation of concrete. (CO1)
- Q.34 Explain pile driving by hammer. (CO2)
- Q.35 Write any five problems faced during pilling process. (CO2)

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Define piles and explain various types of piles. (CO2)
- Q.37 What is anchorage slip? How do you compute the loss of stress due to anchorage slip? (CO1)
- Q.38 Explain the process of pre-tensioned system with diagram. (CO1)

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5th Sem / Civil, Constr. Mgmt. Subject:- Pre-Stressed Concrete

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Small sections are to be constructed in this system:- (CO1)
- a) Post tensioned b) Pre-tensioned
c) Both d) None of these
- Q.2 On the basis of material and composition, which is not the classification of piles (CO2)
- a) Concrete piles b) Composites piles
c) Timber piles d) Plastic piles
- Q.3 Pre tensioning is used for:- (CO1)
- a) Poles b) Bridges
c) Dams d) Plastic piles
- Q.4 The overall diameter of a cable or strand is from:- (CO1)
- a) 5 to 10 mm b) 7 to 17 mm
c) 11 to 15 mm d) 9 to 14 mm

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Q.5 In post tensioning method, class of pre-stress is less (CO1)

- a) 18% b) 10%
- c) 12% d) 15%

Q.6 In circular pre-stressing, the member may be pre-stressed by overlapping. (CO1)

- a) Anchorages b) Ridges
- c) Tendons d) Bars

Q.7 Based on the function, piles can be classified into _____ types. (CO1)

- a) 4 b) 6
- c) 8 d) 3

Q.8 In post tensioning, the concrete units are cast by (CO1)

- a) Ducts b) jacks
- c) Anchorages d) Wedges

Q.9 Which of the following piles is used to compact loose granular soil? (CO2)

- a) Friction piles b) End bearing piles
- c) Compaction piles d) Tension piles

Q.10 The piles that are used for protecting structures from ships and floating object is

- a) Anchor piles b) Fender piles
- c) Compaction piles d) Batter piles

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 What do you mean by strand? (CO1)

Q.12 Piles are generally driven in groups to provide foundations (True/False) (CO2)

Q.13 The steel used in pre-stressed Concrete is called as _____. (CO1)

Q.14 The piles having one or more bulbs are called _____. (CO1)

Q.15 The deflection of a beam with parabolic tendons is _____. (CO1)

Q.16 Define Anchorage. (CO1)

Q.17 In which system, high tensile bars (Tendons) threaded at the ends? (CO1)

Q.18 Loss due to relaxation of stress is (CO1)

Q.19 In pre-tensioned members, reinforcement is tensioned the concrete has fully hardened. (CO1)

Q.20 The grade of concrete for pre-stressed members should be in the range of _____. (CO1)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 What are advantages of pre-stress concrete as compared with R.C.C? (CO1)

Q.22 What are the factors influencing the shrinkage of concrete? (CO1)

Q.23 Discuss loss of stress due to friction. (CO1)

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